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For the 'engineer' in you

Power analyser and power meter: What's the difference?

Power meters are instruments that measure features like peak power, average power and peak-to-average ratio of RF and microwave devices with high accuracy. These are also used to calibrate other instruments and probes. These provide the user with information of power, but cannot determine the region in the frequency spectrum where the device under test is transmitting that power. This means that, even if a device passes the power meter test, it may still be transmitting at the wrong frequency. These instruments are generally used in the communication domain to measure output power of mobile phones, base stations, radar systems, etc. A high-end power meter usually has an external power sensor element, and various sensor technologies are used for this purpose.

Power analysers measure several parameters like direct current (DC), alternating current (AC), DC and AC voltage, intensity, phase rotation, apparent power, effective power, power factor and harmonics. These perform frequency domain analyses and provide both power and transmission frequency information. This helps to overcome false readings that may arise in power meters. Typical application fields include inverters, transformers, power supplies, motor drivers, lighting systems, consumer appliances and other power-conversion devices.

Power-quality measurement using oscilloscopes

Digital oscilloscopes running powerful application software for power-quality measurements can now replace traditional power meters and harmonic analysers. There are benefits of using an oscilloscope over the older toolset. While conventional power meters can overlook signal details due to their relatively slow response time, the high sampling rates of modern oscilloscopes let capture fast-changing events with great resolution. Moreover, an oscilloscope's record length is sufficient to acquire an integral number of cycles, even at a very high sampling resolution.

"Software tools let the user speed-up measurement procedures and minimise the set-up time," says Naresh Narasimhan, country marketing manager of Tektronix India. Most power-quality measurements can be automated by full-featured power-measurement software running on the oscilloscope itself, thereby performing lengthy procedures in seconds. By reducing the number of manual calculations, it acts as a very versatile and efficient power meter. "Also, the oscilloscope probes assist in safe, reliable power measurements," he adds.

Integrated instruments. The general trend of integrating multiple instruments into one system is also seen in the power segment. Latest test systems combine instruments like oscilloscopes, power supplies, multimeters, signal generators and data loggers into a single system. High-definition oscilloscope (HDO) with power-analyser function introduced by Teledyne LeCroy, PX8000 power scopes by Yokogawa and advanced power system (APS) family by Keysight are examples.

Buying a power supply

A variety of AC and DC power sources are available in the market today, ranging from basic to complex models, with a wide range of available capabilities. Given below are a few factors to consider while buying a power supply for your applications.

V & P ratings. The first thing to consider while buying a new power supply is the voltage (V), current (I) and power

(P) ratings required for your target application. Voltage of the supply should neither be too high nor too low as compared to the desired voltage value for the system you want to power. The type of voltage (AC or DC) is also very important. Another requirement is that the current rating of the power supply should be equal to (or greater than) the current necessary to run the system.

Programming capabilities and interfaces. Programmable power supplies allow the user to remotely control its operation through programming via analogue DC signal or digital interfaces like LXI, GPIB (or IEEE-488), Modbus TCP/RCU, RS-232, RS-485 and USB.

Protection. Having integrated power-protection measures help safeguard the instrument, and is very critical while testing costly systems. Over-voltage protection (OVP) and under-voltage protection (UVP) shuts down the supply when outputs are above (OVP) or below (UVP) the predefined trigger point. Over-current protec-